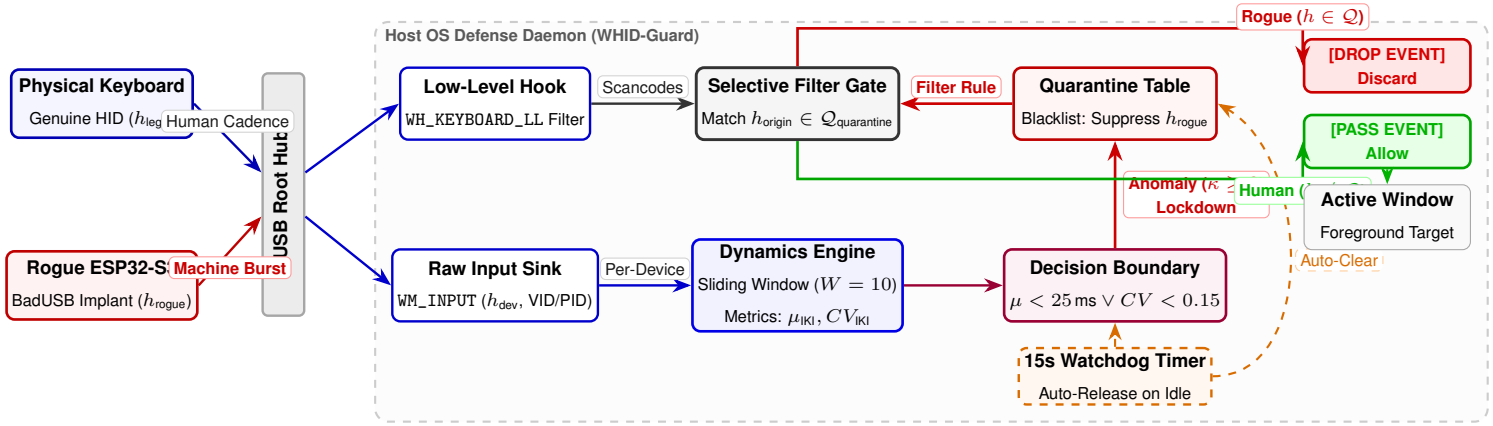

Host OS Defense Daemon (WHID-Guard)
Physical Keyboard
Genuine HID (h_{leg} Human Cadence
Rogue ESP32-S
Machine Burst
BadUSB Implant (h_{rogue})
USB Root Hub
Low-Level Hook
WH_KEYBOARD_LL Filter
Scancodes
Raw Input Sink
WM_INPUT (h_{dev}, VID/PID)
Per-Device
Selective Filter Gate
Match $h_{origin} \in \mathcal{Q}_{quarantine}$
Filter Rule
Dynamics Engine
Sliding Window ($W = 10$)
Metrics: μ_{IKI}, CV_{IKI}
Quarantine Table
Blacklist: Suppress h_{rogue}
Decision Boundary
$\mu < 25\,\text{ms} \vee CV < 0.15$
15s Watchdog Timer
Auto-Release on Idle
Rogue ($h \in \mathcal{Q}$)
[DROP EVENT]
Discard
[PASS EVENT]
Allow
Anomaly (κ
Lockdown
Human (
Active Window
Foreground Target
Auto-Clear